

Strogatz Part I: 1-dim. Flows

- 2. Flows on a line $\dot{x} = f(x), x \in \mathbb{R}$
- 3. Bifurcations $\dot{x} = f(x; \lambda), \lambda = \text{parameter}$
- 4. Flows on a circle $\dot{\theta} = f(\theta), \theta \in S^1$

Dynamical system - evolution rule that determines trajectories as a function of a single parameter (time) on a set of states, the phase space.

For flows, $t \in \mathbb{R}$ is parameter

$\dot{x} = f(x), x \in \mathbb{R} \leftarrow$ phase space is all reals, the allowed states

$x(0) = x_0 = \text{initial condition}$

$x(t; x_0) = \text{trajectory through } x_0 \text{ at } t=0$
parameterized by $t \in J$, some interval containing $t=0$, where

$$\frac{dx(t; x_0)}{dt} = f(x(t; x_0))$$

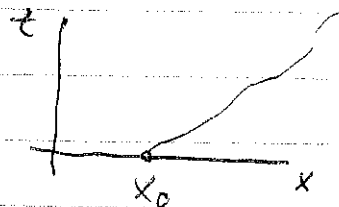
$$x(0; x_0) = x_0$$

$x \in \mathbb{R} \rightarrow$ example of "separable, first-order" eqn.
potentially solvable analytically

$$\int_{x_0}^t \frac{1}{f(x)} \frac{dx}{dt'} dt' = \int_0^t 1 dt = t$$

$$t = \int_{x_0}^x \frac{1}{f(x')} dx'$$

$$= t(x); \quad t(x_0) = 0$$



wanted $x(t)$, not $t(x)$, but maybe we can invert it. (Also, should make sure it's a proper integral, etc.)

Concrete example:

$$\frac{dx}{dt} = x^2 - 1$$

as long as $x \neq \pm 1$

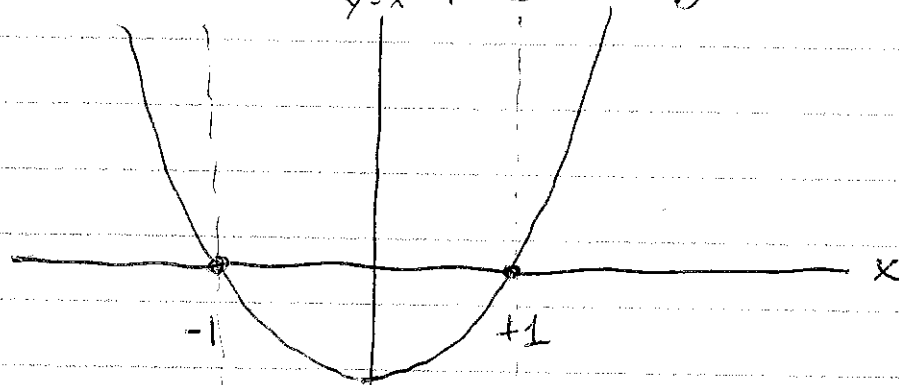
$$\int_{x_0}^x \frac{dx'}{x'^2 - 1} = t$$

$$\frac{1}{x'^2 - 1} = \frac{1}{(x' + 1)(x' - 1)} = \frac{1}{2} \left[\frac{1}{x' - 1} - \frac{1}{x' + 1} \right]$$

$$2t = \ln \left| \frac{x' - 1}{x' + 1} \right|_{x_0}^x \quad x' \neq \pm 1$$

can solve it in various cases explicitly to obtain $x(t)$

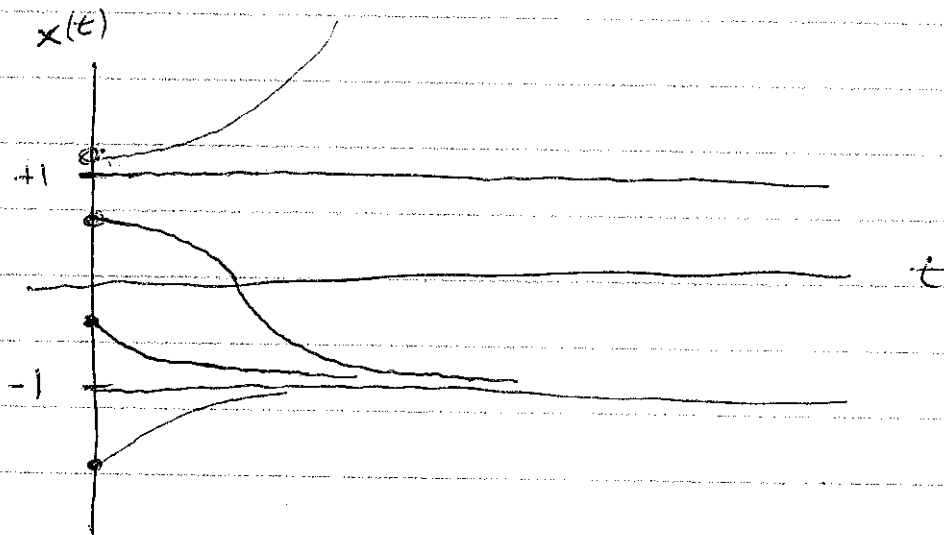
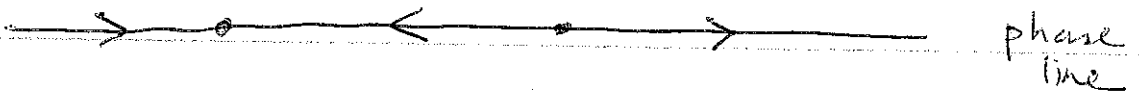
Alternatively, we could obtain solns. qualitatively, based off of just graphing $\dot{x} = x^2 - 1$



$\frac{dx}{dt} > 0$
x increasing

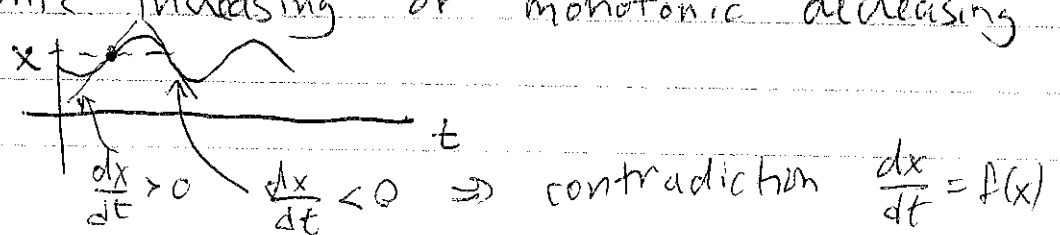
$\frac{dx}{dt} < 0$
x decreasing

$\frac{dx}{dt} > 0$
(x increasing)

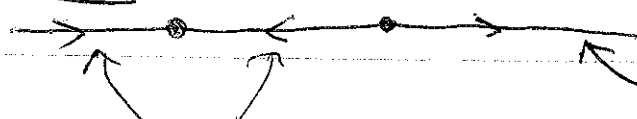


Observations:

- ① Solns. cannot oscillate - if they are non-stationary, then they are either monotonic increasing or monotonic decreasing

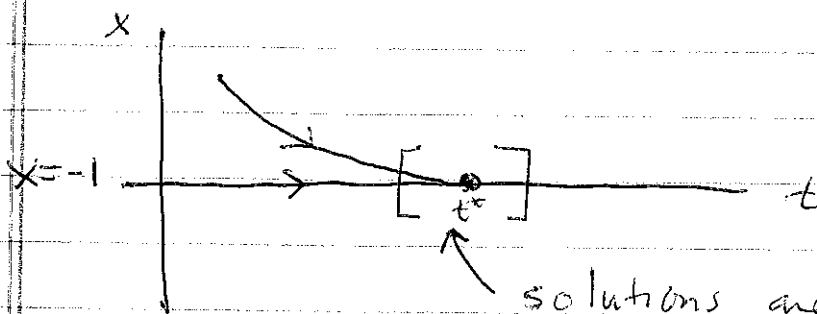


Questions:



can these
get to $x = -1$
in a finite time?

can this diverge
in a finite
time?



solutions are not unique
on interval containing $t = t^*$

Thm. from ODEs:

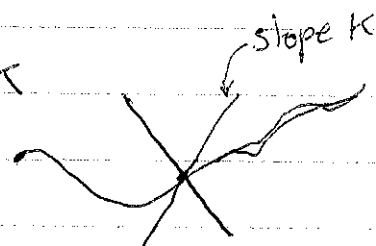
Suppose $f(x)$ is Lipschitz continuous^{*} for all x_1, x_2 in some interval $[x_0 - \delta, x_0 + \delta]$.
Then a unique soln. of $\dot{x} = f(x)$, $x(t_0) = x_0$
exists for all t in some interval
 $t \in (t_0 - \epsilon, t_0 + \epsilon)$.

* Lipschitz continuous: there exists a constant K s.t.

$$|f(x_2) - f(x_1)| \leq K |x_2 - x_1|$$

ie. $\left| \frac{f(x_2) - f(x_1)}{(x_2 - x_1)} \right| \leq K$

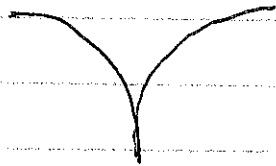
Function
lies inside
the double
cone



Continuous, differentiable functions are all Lipschitz continuous, but a function can be Lipschitz continuous, without being differentiable

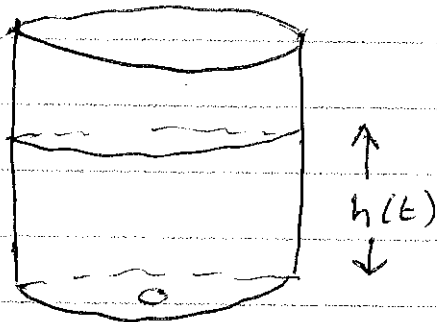


← Lipschitz continuous



← not Lipschitz continuous.

Leaky bucket example: (Strogatz 2.5.6)



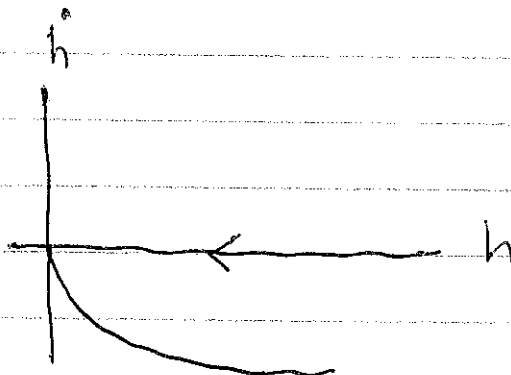
$$\frac{dh}{dt} = -C\sqrt{h}$$

$$C = \sqrt{2g} \left(\frac{a}{A} \right) > 0$$

$$g = 9.8 \text{ m/s}^2$$

a = area of hole

A = cross-sectional area of bucket



Lipschitz continuous for $h > 0$, but not at $h = 0$.

$$h' = -C\sqrt{h}$$

$$h(0) = h_0$$

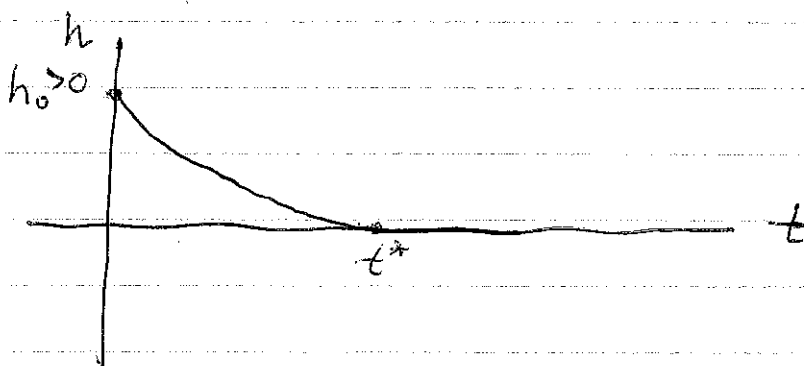
$$\int_{h_0}^h \frac{1}{\sqrt{h'}} dh' = -Ct$$

$$2\sqrt{h'} \Big|_{h_0}^h = 2(\sqrt{h} - \sqrt{h_0}) = -Ct$$

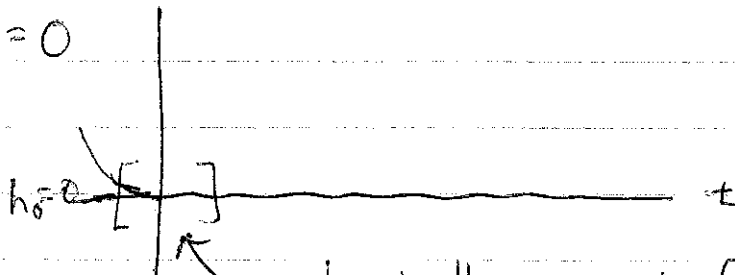
$$\sqrt{h} = \sqrt{h_0} - \frac{C}{2}t$$

$$\text{for } t < t^* = \frac{2\sqrt{h_0}}{C}$$

$$h(t) = \begin{cases} (\sqrt{h_0} - \frac{C}{2}t)^2 & t < t^* \\ 0 & t \geq t^* \end{cases}$$



if $h_0 = 0$



physically meaningful,
non-unique solns.

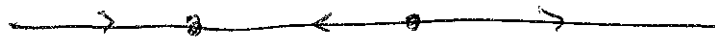
Much of what we'll do is for functions $f(x)$ that are at least C^1 (continuous with continuous first derivative) in which case soln. to

$$\dot{x} = f(x)$$

$$x(0) = x_0$$

is unique on neighborhood of $t=0$, ~~for~~

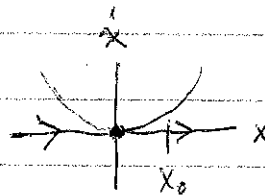
Remaining question



can these trajectories diverge in finite time?

In fact they do.

$$\frac{dx}{dt} = x^2$$



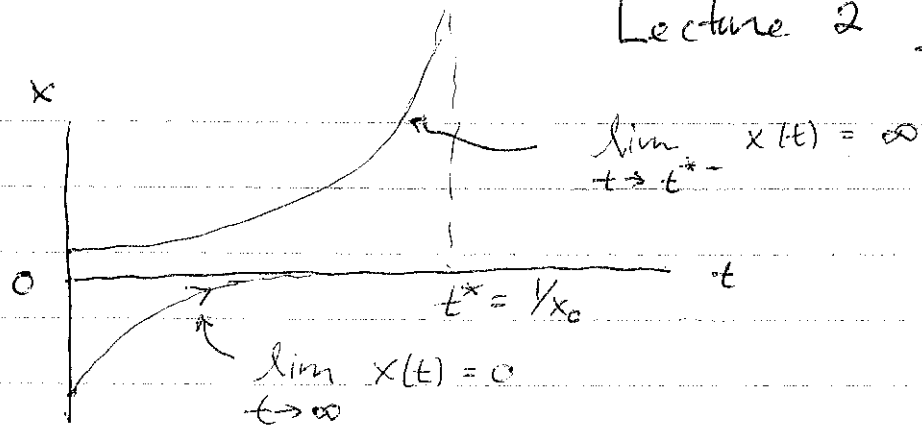
$$\int_{x_0}^x \frac{dx'}{x'^2} = -\frac{1}{x} + \frac{1}{x_0} = t$$

then, $\frac{1}{x} = \frac{1}{x_0} - t = \frac{1 - tx_0}{x_0}$

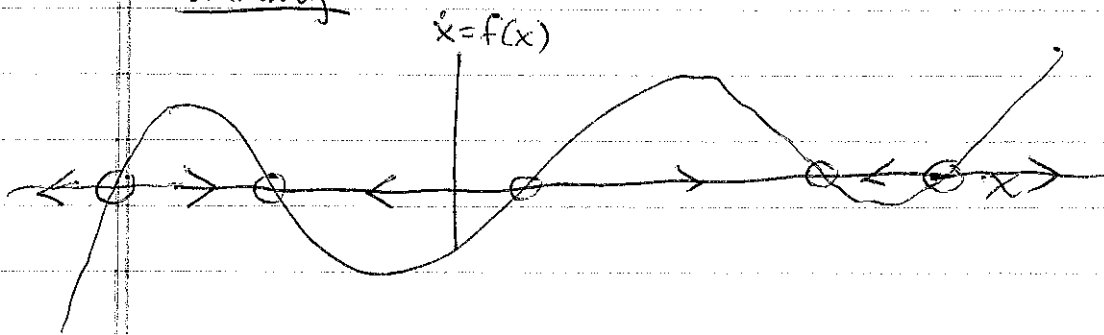
$$x = \frac{x_0}{1 - tx_0}$$

diverges at $t = \frac{1}{x_0} > 0$

Lecture 2 p.8



Summary:



Solns do one of 3 things

- (1) diverge monotonically to $\pm \infty$
- (2) stay put at a point x^* where $f(x^*) = 0$
- (3) asymptotically & monotonically approach $x = x^*$ as $t \rightarrow \infty$